

Part A — Theory: Unsupervised Learning and Clustering Analysis

Clustering algorithms, evaluation metrics, segmentation strategy, and a real-world case study

1. Introduction to Unsupervised Learning and Clustering

What is Unsupervised Learning?

Unsupervised learning is a branch of machine learning in which the algorithm is trained on data that has no human-annotated ground-truth labels. Rather than learning to map input features to a known target value, the goal is to uncover hidden structure, patterns, densities, and relationships that already exist within the feature space itself.

How It Differs from Regression and Classification

Supervised learning (regression and classification) depends on a target vector during training. Classification assigns inputs to a discrete set of class labels — for example, deciding whether a credit card transaction is fraudulent or legitimate — while regression predicts a continuous numerical value, such as tomorrow's temperature. In both cases, the model adjusts its parameters to minimize the gap between its predictions and the true labels.

Unsupervised learning (clustering) operates without any target vector. There is no "correct answer" to check predictions against; instead, the model groups observations or reduces dimensionality based purely on the intrinsic similarities, spatial distances, or density patterns among the input features.

Real-Life Examples

A clustering example is an e-commerce company analyzing raw transaction history, browsing frequency, and geographic location to group anonymous shoppers into behavioral tiers — such as "high-income tech buyers" versus "bargain weekend shoppers" — for targeted promotions. A supervised learning example is a medical model that uses patient biometrics such as blood pressure, age, and insulin levels, paired with confirmed laboratory diagnoses, to predict whether a new patient has diabetes.

2. Clustering Algorithms

A. K-Means Clustering

K-Means is a partition-based algorithm that divides a dataset into K distinct, non-overlapping groups. It begins by placing K random centroids in the feature space, then repeats two steps until the centroids stop moving: each point is assigned to its nearest centroid (typically using Euclidean distance), and each centroid is then recalculated as the mean position of all points assigned to it.

Real-World Use Case: Identifying optimal cell tower placements across a city by grouping geographic user locations to maximize network coverage.

Advantages: Computationally efficient, scales well to large datasets, and is straightforward to implement and understand.

Limitations: Requires the number of clusters (K) to be specified in advance, is sensitive to the initial placement of centroids, and struggles with clusters that are non-spherical or vary greatly in size.

B. Hierarchical Clustering

Hierarchical clustering builds a tree of nested groupings, known as a dendrogram, without requiring a predefined number of clusters. Its agglomerative (bottom-up) form starts with every data point as its own individual cluster, then repeatedly merges the two closest clusters until all points belong to a single master cluster. The resulting tree can later be cut at any height to yield the desired number of groups. Distance between clusters can be defined using single linkage (nearest points), complete linkage (farthest points), or Ward's linkage (minimizing within-cluster variance).

Real-World Use Case: Constructing evolutionary family trees in biology (phylogenetics) by grouping organisms according to genetic similarity.

Advantages: Produces an interpretable visual tree structure and does not require the number of clusters to be chosen in advance.

Limitations: Computationally expensive, with complexity that grows roughly cubically with the number of data points, making it impractical for very large datasets.

C. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

DBSCAN groups points based on density, treating clusters as continuous regions of closely packed observations separated by regions of low density. Two parameters control its behavior: epsilon (ϵ), the radius searched around each point, and MinSamples, the minimum number of neighboring points required to form a dense region. Points are classified as core points (with at least MinSamples neighbors within ϵ), border points (near a core point but without enough neighbors of their own), or noise points (isolated observations that satisfy neither condition). Clusters grow by connecting adjacent core points, while noise points remain unclustered.

Real-World Use Case: Detecting anomalous or fraudulent credit card transactions by identifying outliers that fall far outside normal spending clusters.

Advantages: Naturally discovers clusters of irregular shape, requires no predefined number of clusters, and automatically isolates noise and outliers.

Limitations: Performs poorly on datasets with widely varying densities, and its results depend heavily on selecting appropriate values for ϵ and MinSamples.

3. Clustering Metrics

Elbow Method (Sum of Squared Errors): Measures the total squared distance between each point and its assigned centroid, indicating how compact the clusters are. As K increases, SSE naturally decreases; the goal is to identify the "elbow" point where further increases in K yield diminishing improvement.

Silhouette Score: Evaluates how well-separated clusters are by comparing each point's average distance to other points in its own cluster against its average distance to points in the nearest neighboring cluster. The resulting score ranges from -1 to +1, with higher values indicating cleaner separation.

Davies-Bouldin Index: Compares the internal scatter of each cluster to the separation between clusters, averaged across all clusters. Lower values indicate a better, more clearly separated clustering solution.

Table 1. Comparison of clustering evaluation metrics.

Metric	What It Measures	Ideal Value	Sensitivity to
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			Shape/Imbalance
Elbow Method (SSE)	Internal cluster compactness and rigidity.	A distinct "elbow" where SSE flattens out.	Highly sensitive to non-spherical shapes; favors compact, round clusters.
Silhouette Score	How close a point is to its own cluster versus other clusters.	Closer to +1.0.	Reliable for general shapes, but can penalize complex density-based geometries.
Davies–Bouldin Index	Ratio of cluster tightness to distance between clusters.	Lowest possible score, closer to 0.0.	Tends to favor spherical arrangements; less reliable with complex geometries.

4. Choosing K and Interpreting Segments

Selecting the Number of Clusters (K) for K-Means

Selecting an appropriate K combines quantitative evidence with practical business judgment. First, the SSE is plotted across a range of K values (for example, 1 through 10), and a distinct bend, or elbow, is identified where additional clusters no longer meaningfully reduce SSE. Second, the average Silhouette Score is computed across the same range, and the K that maximizes this score suggests the cleanest separation. Finally, a domain check confirms that the resulting groups make practical business sense — each segment should be distinct and large enough to justify a targeted strategy.

Interpreting Spend Combinations (Wholesale Distributor Context)

A cluster showing high spend on fresh produce and milk typically represents clients focused on perishable goods — such as restaurants, fresh food cafés, or local grocery stores that restock frequently to maintain product freshness. A cluster showing high spend on groceries, detergents, and paper products typically reflects bulk, non-perishable purchasing behavior associated with large supermarkets, corporate supply chains, department stores, or hospitality venues that stock goods with a long shelf life.

Why Channel and Region Are Excluded From Clustering Features

Categorical variables such as Channel (for example, Retail versus Horeca) and Region (for example, Lisbon versus Oporto) are excluded from the clustering feature space for three main reasons. First, mathematical incompatibility: clustering algorithms such as K-Means compute distances in Euclidean space, and categorical codes have no meaningful numerical distance — Region 3 is not "three times greater" than Region 1, so including them distorts the distance calculation. Second, preventing latent grouping bias: including these variables risks having the algorithm simply reproduce the existing categories rather than uncovering new, previously hidden behavioral patterns. Third, keeping these variables aside allows for post-hoc validation, where the discovered clusters can later be compared against Channel and Region to check whether certain buying behaviors naturally align with particular regions or business types.

5. Real-World Case Study

Project Goal: A major European multi-brand fashion retailer sought to move from generic, seasonal email campaigns toward hyper-personalized marketing by automating customer segmentation.

Data Used: Behavioral data from 45,000 active customer accounts over a twelve-month period, including Recency, Frequency, and Monetary value (RFM metrics), the percentage of items purchased on discount, and product category return rates.

Clustering Method Applied: K-Means clustering combined with Principal Component Analysis (PCA) for initial dimensionality reduction. The optimal number of clusters was determined to be $K = 4$ using a combination of the elbow method and silhouette validation.

Key Results and Strategic Insights: The model identified four distinct customer segments: "Brand Loyalists," characterized by high frequency and spend with low discount sensitivity and low returns; "Deal Hunters," who spend less overall but respond strongly to promotions; "High-Volume Returners," who purchase and spend heavily but return more than 60 percent of items, suggesting wardrobe-testing behavior; and "Lapsed Occasional Buyers," marked by long absences and low overall spend. By tailoring marketing strategies to each segment — for instance, sending exclusive previews to Brand Loyalists and targeted clearance offers to Deal Hunters — the retailer achieved a 34 percent increase in email click-through rate and a 12 percent reduction in logistics costs associated with returns within two financial quarters.

References

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